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Assessing personality using zero-shot generative AI scoring of brief open-ended text

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Contemporary personality assessment relies heavily on psychometric scales, which offer efficiency but risk oversimplifying the rich and contextual nature of personality. Recognizing these limitations, this study explores the use of commercially available generative large language models (LLMs), such as ChatGPT, Claude and so on, to assess personality traits from open-ended qualitative narratives. Across two distinct samples and methodologies (spontaneous streams of thought and daily video diaries), we used seven commercial, generative LLMs to score Big-Five personality traits, achieving convergence with self-report measures comparable to or exceeding established benchmarks (for example, self–other agreement, ecological momentary assessment, and bespoke machine learning models). Although results differed across different LLMs, we found that using the average LLM score across models provided the strongest agreement with self-report. Further, LLM-generated trait scores also demonstrated predictive validity regarding daily behaviours and mental health outcomes. This LLM-based approach achieved quantitative rigour based on qualitative data and is easily accessible without specialized training. Importantly, our findings also reaffirm that personality is expressed ubiquitously, in that it is carried in the stream of our thoughts and is woven into the fabric of our daily lives. These results encourage broader adoption of generative LLMs for psychological assessment and—given the new generation of tools—stress the value of idiographic narratives as reliable sources of psychological insight.

Contemporary models of personality, such as the Big-Five and Five-Factor models, have their origins in the lexical hypothesis, which posits that the most important attributes of personality should be sedimented in the everyday language used to describe people (see ref. 1 for a review). The underlying structure of personality can then be revealed by systematically sifting through ratings of trait descriptive terms for patterns (for example, with factor analysis) to establish fundamental attributes along which any individual could be quantified. The Big-Five and similar models (that is, the various Big-Few models) were the result of this process and now enjoy widespread acceptance in the field. This approach to personality science led to the development of psychometric scales designed to reliably and validly measure personality traits through standardized questionnaires. Such scales offer numerous advantages: they are efficient, easy to administer, psychometrically robust and facilitate comparisons across individuals and contexts.

Despite these benefits, over-reliance on psychometric scales risks constraining our understanding of personality, and the human condition more broadly, through its dependence, if not reification, of specific scales that are not always well suited to the complexity and contextual richness of an individual's life experiences. The inherent nomothetic nature of psychometric assessment filters nuanced

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individual differences through brief, fixed sets of questions, potentially neglecting important qualitative information[2,3]. Scale development often subordinates breadth to internal consistency[4], and strong assumptions must be made about whether and how the experiences that are queried apply to someone's life. This narrows the aperture of our scientific lens and constrains the information gathered. Only the questions asked can be answered, and all questions are asked of everyone whether they are relevant or not.

Recognizing these limitations, personality research has increasingly sought alternative assessment methods that transcend the constraints of traditional questionnaires. Some of these methods still rely on traditional question-based scales but dramatically increase the breadth of content at the item level[5], treat individual items as constructs[6] or use ecological momentary assessment to conduct assessments across contexts in daily life[7]. Other approaches do not rely on questionnaires, turning instead to concrete behavioural data, for instance, passive sensing from smartphone sensors[8] and social media 'likes'[9].

Among the most promising developments has been a return to studying personality as reflected in language as it is expressed naturalistically in open-ended form, rather than in lists of trait descriptive adjectives. Open-ended, qualitative data offer considerable potential to deepen our understanding of personality by capturing an individual's experiences in a rich, nuanced and ecologically valid form. Unlike structured questionnaires, open-ended narratives provide greater fidelity to personal experiences because they allow individuals to express themselves freely, highlighting information that is most relevant and meaningful to them. This approach can uncover insights that researchers might not anticipate and contextualize responses in ways that structured scales often cannot. Moreover, open-ended data are particularly valuable for capturing experiences dynamically, as they unfold over time. However, despite these benefits, qualitative data remain relatively underutilized in personality assessment owing to the substantial practical challenges involved[10]. The process of transcribing, coding and converting qualitative narratives into actionable quantitative data imposes appreciable burdens. As a result, the efficiency and depth gained from open-ended data collection are often diminished or lost when researchers attempt to systematically analyse and score these data.

A method that retains the qualitative depth of open-ended responses while enabling reliable, efficient quantitative scoring would therefore represent a transformative advancement for personality science, blending the strengths of both qualitative and psychometric traditions in an advanced version of the lexical hypothesis. This could be achieved using natural language processing, specifically large language models (LLMs), to assess personality from naturally produced, open-ended text[11]. Initial results are promising. For instance, Park et al.[12] used an open vocabulary approach with over 66,000 participants' Twitter posts to derive language features that indicate Big-Five trait levels, achieving strong convergent validity (mean $r = 0.38$) and test–retest correlations comparable to self-report measures ($r_s = 0.65$–0.71). Kulkarni et al.[13] demonstrated that novel personality dimensions could be uncovered from Facebook data that predicted important outcomes better than traditional, close-ended questionnaire measures of personality in many instances.

However, these impressive results have relied on bespoke LLMs that require large training datasets and deep technical expertise to implement. The substantial resources and expertise required create barriers that make these approaches impractical for most researchers. These challenges may not be insurmountable for well-resourced labs, but they are a heavy climb for all. By contrast, commercial generative artificial intelligence (AI), exemplified by widely available models such as ChatGPT and Claude, offers an exciting opportunity to democratize this advanced approach. These generative AI systems have the potential to reliably score personality traits directly from open-ended qualitative data through 'zero-shot' approaches (that is, requiring no task-specific model training) owing to their extensive semantic understanding and contextual embedding capabilities. Leveraging pre-trained, commercially available generative AI in personality assessment could thus blend the qualitative richness and contextual depth of open-ended narratives with the quantitative rigour and efficiency traditionally associated with psychometric scales.

Some work has already started testing this out in a limited way. For example, Rathje and colleagues[14] found that ChatGPT (3.5 Turbo, 4 and 4 Turbo) outperformed ($r_s = 0.59$–0.77) dictionary-based approaches ($r_s = 0.20$–0.30) at detecting targets such as sentiment and discrete emotions from tweets and headlines. Peters and Matz[15] similarly used the MyPersonality[12] dataset to examine whether ChatGPT (3.5 and 4) could infer personality traits from zero-shot ratings of social media (Facebook) posts (average $r = 0.29$).

Several studies have also used ChatGPT for text analysis as it relates to depression, a near-neighbour construct to personality[16]. For instance, Hur et al.[17] showed that ChatGPT (3.5 and 4o) could be used to extract sentiment valence (that is, positivity versus negativity) from open-ended question responses from neutral prompts based on a popular depression questionnaire (Patient Health Questionnaire-9). These sentiment scores were associated with current mood, predicted depression change over 3 weeks and outperformed simple word counting software. Another study by Shin et al.[18] used ChatGPT (3.5 and 4o) to classify 91 individuals as depressed or not based on brief, semi-structured daily diaries with impressive accuracy (>.8). In addition, GPT-4o ratings of depression symptoms (that is, Patient Health Questionnaire-9 items) from participant narratives of their own depression correlated as or more strongly with self-reports and expert ratings of those items than did self-reports and expert ratings[19]. Moreover, the same study showed that ChatGPT was capable of inferring presence of symptoms that were not explicitly mentioned in the narratives, attesting to its mapping of the depression construct.

Each of the reviewed studies involved the use of qualitative, open-ended text data, but with several notable qualifications. The depression studies used text that was open-ended but highly constrained, with participant prompts limited to eliciting narratives of depression or open-ended questions based on depression questionnaire items. Similarly, Rathje et al.'s[14] study, though impressive in its scope of stimuli, was limited in what was rated, largely focusing on sentiment valence. Peters and Matz[15] expanded this to the Big Five, which is our goal here as well, with mostly encouraging results. At the same time, limiting their analysis to social media posts is a specific context that may not generalize to the types of data that many investigators might want to study and highly public posts might be uniquely vulnerable to self-presentation biases. Notably, Facebook posts may not adequately reflect the content of an individual's inner thoughts and narratives, only what Facebook users are willing to share publicly. The degree to which traits are expressed outwardly influences their ratability[20]. For instance, neuroticism often has lower self–other agreement, given that it involves internal emotional states, and that was the case for self-LLM agreement in Peters and Matz's analyses. Facebook posts might similarly be biased against including mundane behaviour, such as planning, cleaning, organizing and such that reflects conscientiousness. Consistent with this possibility, their LLMs had lowest agreement with self-reports of conscientiousness.

Apart from Ganesan et al.[19], none of the studies used an LLM-scoring method that allowed for investigations of construct coherence. There was also no evaluation of broader patterns of correlations with external criteria, instead prioritizing the study of agreement with self- or clinician ratings of the same constructs scored by the LLMs. As a result, these previous studies lack any real investigation of internal consistency and construct validity (that is, nomological networks) of the LLM ratings, which are the hallmark of personality science and the psychometric approach more broadly. Finally, all these studies

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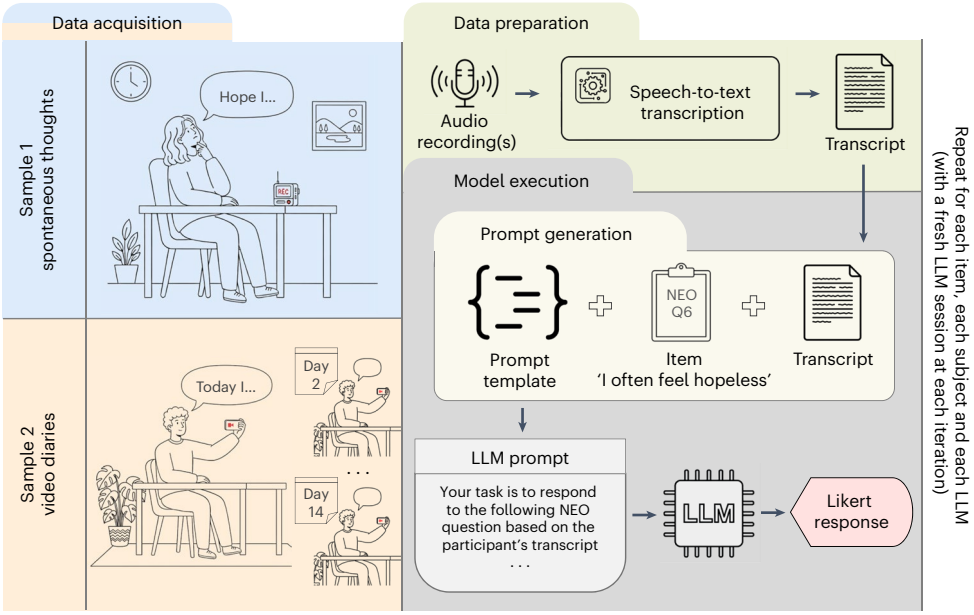

Fig. 1 | Diagram of procedures across samples. Left side panels depict the different methods of data acquisition in each sample. Sample 1 (blue panel) was recorded in the laboratory during a single session. Sample 2 (peach panel) was collected in the naturalistic settings of participant's life over multiple days. Both samples followed nearly identical procedures for transcribing text, prompting the large language model and scoring the individual items as depicted in the panels for data preparation and model execution.

were limited to using a single company's LLMs—OpenAI's ChatGPTs. Thus, as impressive as the ChatGPT performance is in these zero-shot studies has been, their narrow focus leaves open questions of whether, and, if so, how well the many other, commercially available pre-trained transformer models perform with scoring wide ranges of constructs from less constrained qualitative data.

The current study

What we propose here is a test of generative AI's capability for quantifying the Big-Five/Five-Factor model traits: extraversion, agreeableness, conscientiousness, neuroticism and openness to experience from unconstrained, qualitative text data. Not only are the Big-Five traits the natural target for evaluating whether this approach is feasible for advancing personality science, but also they are broadly important for the behavioural sciences writ large, in that they cover wide swaths of human mental and behavioural functioning, including emotional/affective, interpersonal, cognitive and behavioural domains. Indeed, recent studies show that most popular trait scales can be organized within this framework[21]. Moreover, the Big Five have strong associations with psychiatric diagnoses[22] and are useful in employment selection contexts[23], as well as being associated with most of life's most important functional outcomes[24]. In addition, they are arguably the prime example of constructs that have emerged from the lexical hypothesis and the psychometric tradition[1,25]. Thus, as targets, they offer an exceptional test of AI's capability for extracting and quantifying critical psychological content and abstracting beyond the content contained in a specific unstructured and unconstrained report.

Furthermore, we use two very different datasets of open-ended unconstrained reports, collected for distinct purposes in dissimilar contexts to illustrate the flexibility and generalizability of this approach. First, we use transcripts of narratives that were recorded from 60 individuals who came to the lab and verbalized their spontaneous stream of thoughts over 30 min (average word count = 3,097, SD = 1,195, range = 361–5,593) in response to the prompt, 'The only thing you need to do is let your spontaneous thoughts unfold like they would in the situations above (like the long bus ride or the long shower) except this time you say these thoughts out loud'. Second, we use transcripts from brief daily video diaries, recorded nightly from participants (N = 108) over 15–21 days (day N = 1,513, average word count = 1,651, SD = 976, range = 66–3,938) in response to the prompt, 'Please think of the most significant event in your day today. Use the video to briefly (1 min) describe: what happened, who was there, how you behaved, and how you felt during the event'. These each represent open-ended prompts with very little in the way of guidance for what should be reported, one representing spontaneous streams of thoughts as might occur in mind wandering, whereas the other represents repeated brief summaries and reaction to events in one's daily life. In both cases, participants were completely free to choose the content and style of their response. Also, in each sample, participants completed baseline assessments that included traditional self-report measures of Big-Five personality trait inventories.

Results

The general methodological approach was the same in each sample (see Fig. 1 for a depiction); we first transcribed the recorded open-ended responses using a local instantiation of Open AI's Whisper[26] and then submitted the transcripts to multiple generative LLMs (ChatGPT 4.1, ChatGPT 4.1 Mini, Claude 3.7 Sonnet, Gemini 2.5 Flash, Grok 3 Beta, Llama 4 Maverick and Qwen 235B), prompting them to respond to each item of the same personality inventory that the participants had completed as if the LLMs were the participant that generated the open-ended responses (see Methods for details of prompts). The individual items were then scored following each measure's protocol to provide LLM-generated personality trait scores that could be correlated with participant-generated personality trait scores on the same inventories (both samples) and other measures (Sample 2). Because item-level LLM responses were available, we investigated internal consistency of the scales in the data resulting from the LLM scoring. Internal consistency provides information on how well the constructs hang together or cohere as reflected how consistently the indicators are endorsed. In Sample 2, we also collected a wealth of additional variables that allowed us to compare the nomological networks of self- and LLM-generated personality trait scores, including self-report inventories, self-reports of daily behaviour collected using ecological momentary assessment and mental health outcomes (treatment status, internalizing diagnoses and personality disorder severity).

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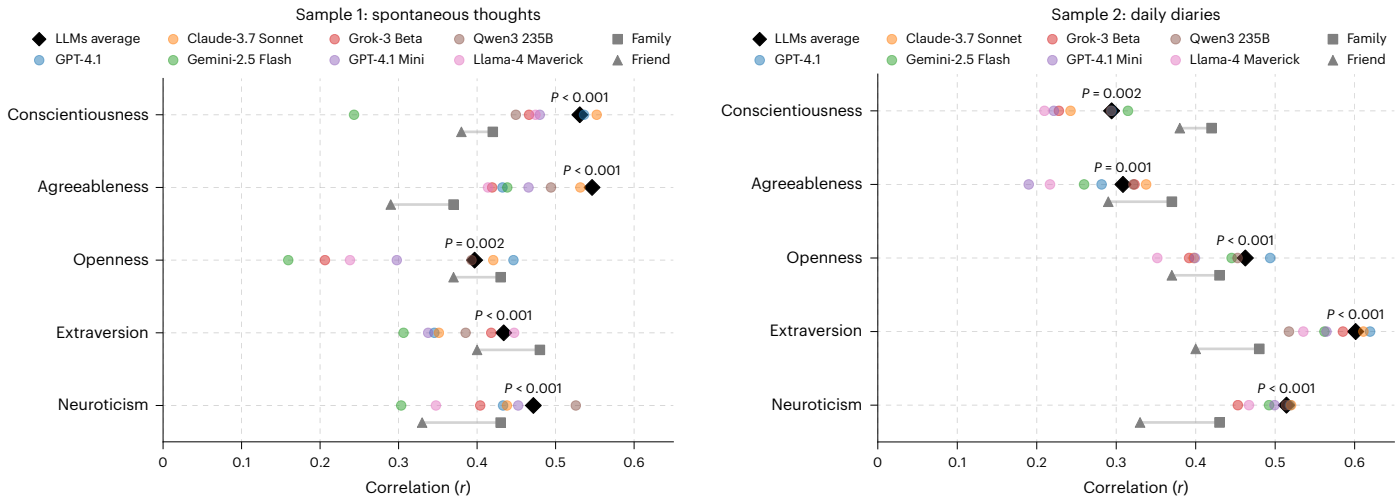

Fig. 2 | Correlations between LLM estimates of personality from stream of thought prompts (Study 1, left panel) and daily diaries (Study 2, right panel) and self-report survey compared with meta-analytic estimates of self–other survey agreement (Connelley and Ones, 2010). For all but conscientiousness assessment based on daily diaries, LLM-based estimates match or exceed typical self–other agreement benchmarks for family and friends (grey boxes and triangles, respectively). Values in the figure represent Pearson correlations (r). In Sample 1 ($N = 60$, df = 58), values of $r \geq 0.42$, $r \geq 0.33$ and $r \geq 0.25$ are significant at $P < 0.001$, $P < 0.01$ and $P < 0.05$, respectively, using two-sided tests. In Sample 2 ($N = 108$, df = 106), all values of $r \geq 0.31$, $r \geq 0.25$ and $r \geq 0.19$ are significant at $P < 0.001$, $P < 0.01$ and $P < 0.05$, respectively, using two-sided tests. No adjustments were made for multiple comparisons.

Detailed descriptions of study procedures, LLM prompts and self-report measures are provided in Methods ('Measures' section). This study was not pre-registered.

Convergent associations between LLM-generated and self-reported trait scores

In Sample 1, participants provided verbalizations of their spontaneous streams of thoughts and completed the 60-item NEO-FFI Personality Inventory[27]. First, we examined convergent correlations of each self-rated personality trait with the corresponding LLM-generated trait scores. We did this for each individual LLM separately as well as average LLM-generated trait scores, which were created by taking the average across all the LLM scores. Results as plotted on the left panel of Fig. 2 (tables of numerical values and correlations among different LLM scores can be found in Supplementary Tables 1 and 2). Figure 2 also includes meta-analytic family/friend ratings[20] for comparison. The magnitude of associations for specific LLMs with self-reported traits varied somewhat both within and across traits (range of $r_s = 0.16–0.55$). In addition, convergent correlations for the average of the LLM scores were among the strongest for all traits (range of $r_s = 0.30–0.53$). This result is consistent with the increased reliability and validity that is observed when using the average of multiple human raters in coding studies[28], reflecting the wisdom of the crowds. The strongest convergence between self-report and LLM ratings were for conscientiousness (average of individual LLM r_s using Fisher r to z transform = 0.46) and agreeableness (average individual LLM $r = 0.46$) and followed by neuroticism (average individual LLM $r = 0.44$). When considering the performance of individual LLMs, the Gemini 2.5 Flash had the lowest average convergent correlation with self-reports across traits (average $r = 0.29$), with the remaining LLM convergent correlation averages ranging from 0.39 to 0.46. Thus, with one exception, the individual LLMs performed very similarly on average. The LLM average had an average convergent correlation with self-report of 0.48 in this sample.

In Sample 2, participants provided nightly video diaries and completed the 120-item International Personality Item Pool-NEO (IPIP-NEO[29]) personality inventory, but otherwise the analytic steps were the same as for Sample 1. The results for convergent correlations between self-rated and LLM-scored traits are presented in the right panel of Fig. 2. In Sample 2, across traits, we found more variable but still consistently significant correlations like in Sample 1. In contrast to Sample 1, the weakest effects were for conscientiousness (average individual LLM $r = 0.26$) and agreeableness (average individual LLM $r = 0.28$), with the strongest effects for extraversion (average individual LLM $r = 0.57$) and neuroticism (average individual LLM $r = 0.49$). Convergence with LLM average scores was stronger than most individual LLMs, like in Sample 1, although these were not typically the strongest effects in this sample. Looking across LLMs in Sample 2, we found correlation averages that ranged from 0.37 to 0.45. This is very similar to Sample 1, with the exception that Gemini 2.5 Flash was no longer an outlier, with an average of $r = 0.42$. The LLM average had an average correlation of 0.44 in this sample. Also of note is that the average of all correlations across all LLMs and traits in each sample was an identical $r = 0.415$. Yet, as is clear in Fig. 1 and the tabulated values in Supplementary Tables 1 and 2, this remarkable consistency masks a great deal of heterogeneity across individual models and traits.

Evaluating the nomological net of LLM-generated trait scores

Sample 2 also affords us the opportunity to examine the nomological nets of LLM-generated trait scores and whether they are similar to the nomological net of self-rated traits. We did this by estimating associations between LLM-generated trait scores and multiple external criterion measures, as well as the associations between self-reported trait scores and the same external measures, and then calculating the correlation between the LLM and self-reported profiles of associations. These resulting profile correlations tell us how similar the LLM and self-rated traits are as constructs more broadly, not just their 1:1 association. For this, we chose 41 external self-report scales measuring interpersonal problems and maladaptive personality features, which include scales covering constructs such as negative affectivity, disinhibition, antagonism and psychoticism. All of these have well-established and differential associations with the Big-Five traits[30] and therefore serve as useful targets for testing the nomological networks of the LLM-generated trait scores. For the self-reports, each individual LLM and LLM average, we calculated each trait's profile of correlations with the external measures and then calculated the correlations of the profiles between self- and LLM-generated trait scores, all of which can be found in Table 1 (see Supplementary Tables 3–11 for the profiles of associations). Here the average of the LLM scores provides a reasonable

Table 1 | Profile correlations among associations between self-reported personality traits and 41 self-reported maladaptive personality trait scales and LLM-scored traits and the same external variables

Model	Openness	Conscientiousness	Extraversion	Agreeableness	Neuroticism
Claude Sonnet	0.69	0.67	0.96	0.72	0.92
Gemini Flash	0.79	0.80	0.95	0.52	0.93
ChatGPT 4o Mini	0.79	0.63	0.95	0.45	0.92
ChatGPT 4o	0.75	0.61	0.97	0.61	0.92
Grok-3 Beta	0.75	0.62	0.97	0.74	0.91
Llama Maverick	0.70	0.61	0.91	0.33	0.90
Qwen3 235B	0.79	0.72	0.94	0.62	0.91
LLMs average	0.78	0.72	0.96	0.62	0.92

Extraversion and neuroticism are most effectively scored by the LLMs, agreeableness the least.

Table 2 | Correlations between the average of LLM-scored traits, self-reported traits and daily life behaviour collected with ecological momentary assessment and mental health outcomes collected via clinical interview

	Extraversion		Agreeableness		Conscientiousness		Neuroticism		Openness	
	LLM	Self	LLM	Self	LLM	Self	LLM	Self	LLM	Self
Positive affect	***0.38***	***0.49***	−0.05	0.13	0.14	***0.37***	***−0.30***	***−0.48***	0.16	**0.20**
Negative affect	***−0.26***	***−0.30***	−0.06	−0.08	***−0.25***	***−0.29***	***0.50***	***0.52***	0.05	0.02
Warm behaviour	0.10	−0.13	***0.30***	0.12	0.12	−0.06	−0.08	0.16	0.17	0.04
Dominant behaviour	0.15	0.16	−0.16	**−0.20**	−0.07	**0.23**	−0.09	−0.15	−0.07	−0.17
Daily stress	−0.11	***−0.28***	0.08	0.07	−0.06	−0.15	***0.34***	***0.29***	0.10	−0.09
Mental health treatment	−0.18	***−0.30***	−0.02	0.17	−0.09	**−0.20**	***0.46***	***0.42***	***0.26***	0.02
Internalizing diagnosis	**−0.23**	***−0.31***	−0.16	**−0.19**	**−0.24**	−0.19	***0.34***	***0.54***	0.04	0.02
Personality disorder	**−0.24**	**−0.23**	***−0.25***	***−0.37***	***−0.37***	***−0.37***	***0.38***	***0.54***	0.06	0.06

Most outcomes are, to some degree, self-reported and therefore associations with average LLM scores are weaker because they reflect cross-method associations and the outcomes. The exception is whether participants received mental health treatment. Note: N=108. Values in the table represent Pearson correlations (r; df=106) for which SEs=0.098, and all values of $r \geq 0.31$, $r \geq 0.25$ and $r \geq 0.19$ are significant at $P<0.001$, $P<0.01$ and $P<0.05$, respectively, using two-sided tests. No adjustments were made for multiple comparisons. All significant effects (that is, at least $P<0.05$) are bolded, $P<0.01$ are italicized, and $P<0.001$ are italicized and underlined. Affect, interpersonal behaviour and stress are averages of ecological momentary assessments. Internalizing diagnoses and global personality disorder severity are all based on clinical interviews.

representation of effect sizes, and we found that the nomological networks were overall quite similar considering the methodological differences (profile r_s = 0.62–0.96). Profile correlations varied across traits with LLM-scored extraversion (profile r for LLM average = 0.96) and neuroticism (profile r for LLM average = 0.92) showing substantially more similar nomological nets to self-reported traits than agreeableness (profile r for LLM average = 0.62) and conscientiousness (profile r for LLM average = 0.72), and openness falling in between (profile r for LLM average = 0.78).

Associations with daily behaviour and common mental health outcomes

We next evaluated whether LLM-generated personality trait scores predict consequential, real-world behaviour and mental health outcomes, namely, day-to-day positive and negative emotions, warm and dominant interpersonal behaviour, daily stress, mental health treatment seeking, and interviewer assessed internalizing spectrum and personality disorder symptoms. Correlations of the average of the LLM-generated trait ratings with each of these variables can be found in Table 2 alongside the self-reported trait associations. The patterns of associations for LLM traits mirrored the self-reported associations, albeit somewhat weaker owing to the lack of shared method variance. LLM-rated extraversion was significantly correlated with higher average positive emotion and lower negative emotion ratings in daily life and negatively associated with interview rated internalizing diagnoses and personality disorder symptoms. Agreeableness was associated with more average warm behaviour and negatively associated with personality disorder symptoms. Conscientiousness was related to lower negative affect, internalizing and personality disorder diagnoses. Neuroticism was correlated with higher average ratings of daily life stress and negative emotions, lower ratings of positive emotions, more treatment seeking, internalizing diagnoses and personality disorder symptoms. Openness was associated with more treatment seeking.

Comparison to other natural language processing approaches

How do our results compare to existing language-based scoring algorithms? We first compared them with the most widely used approach to natural language processing in psychology, Linguistic Inquiry and Word Count (LIWC[31]) software version 15, using all 93 features from its dictionaries to predict the self-reported traits in each sample. To accomplish this, we used linear regression with fivefold holdout sampling and saving the average multiple R value across samples. Results are visualized in Fig. 3 for each sample, with a comparison to the worst- and best-performing LLM for each trait as well as the average LLM association. In all cases, except two (Sample 1 neuroticism and openness), we found that even the weakest LLM-generated trait association was stronger than the LIWC association, often markedly so. In fact, in half of the cases, the LIWC-scored trait correlations were negligible (that is, <0.10). Thus, LLM-generated traits typically far outperform LIWC in our samples.

Next, we scored Big-Five personality traits in each of our samples using an existing state-of-the-art machine learning-based prediction model (MyPersonality; yielding weights on words), which were trained on 12 million social media posts from 65,896 users using self-rated Big Five as ground truth[32]. These models have been shown to be far more powerful at predicting outcomes than LIWC, providing a more stringent benchmark for our approach[11]. As expected, we found that the MyPersonality machine learning model trait estimates correlated

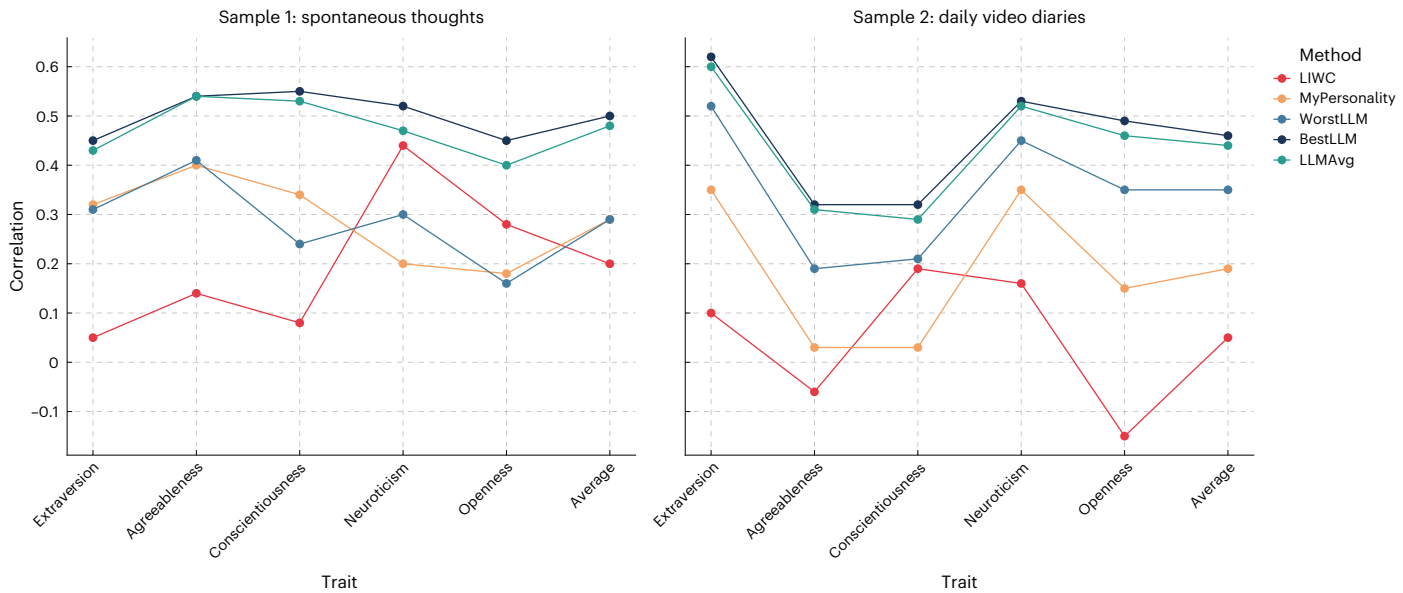

Fig. 3 | Correlations between self-reported personality traits and alternative natural language processing approaches. WorstLLM reflects the lowest observed association between any model and the trait, whereas BestLLM reflects best observed association between any model and the trait. LLMAvg is the average of all LLMs combined into a single score. Associations with LIWC based on version 15 and are the multiple *R* values from regression models predicting traits from all LIWC features from built-in dictionaries and using fivefold cross-validation MyPersonality reflects the use of the MyPersonality model weights.

more strongly on average with self-report traits in both samples when compared with the LIWC correlations. In some specific cases, the MyPersonality model performed more poorly than LIWC (Sample 1 neuroticism and openness, Sample 2 conscientiousness). Compared with the LLM-scored trait associations, in Sample 1, the MyPersonality model performed as well, on average, as the worst-performing generative LLM, but in Sample 2, even the worst LLM associations were higher than those from MyPersonality-scored traits, sometimes by a substantial margin. Given the relatively short texts present in our study datasets, the contextual information that LLMs can consider (in the technical sense of leveraging contextual embeddings) likely has a clear advantage on the purely lexical information (word occurrences) used by the word-level machine learning model.

Unpacking the LLM-generated trait scores

In theory, the ability of LLMs to perform better than prevailing natural language processing methods is because they use richer, contextually embedded information from the narratives to assign scores. But do they? One of the advantages of using generative LLMs for psychological construct scoring is the ability to prompt these models to provide summaries of their rationales for scoring and support their scores with representative statements from the narratives. Although this method may not directly reflect the internal model reasoning, this approach can be used for checking the work of the models; that is, are the resulting quotes and rationales obviously part of the construct intended for scoring? It can also be used to gain insight into the constructs under study, providing insight into questions such as how are these constructs typically manifested in behaviour naturalistically, are there more or less common manifestations of the relevant constructs, and what other behaviours or contexts are often described in tandem.

Table 3 contains example quotes from each sample drawn from summaries of individuals scored as high and low in each trait. What these examples clearly highlight is that individuals make statements that are squarely aligned with the constructs that are being scored. In some cases, statements are quite like items on the standard personality inventories, but in many other cases, statements are both representative of the domain but also highly idiosyncratic, illustrating the LLMs' ability to identify content related to complex psychological constructs from novel stimuli.

Internal consistency of LLM-generated trait scores

Finally, because our method of LLM scoring generates item-level responses, we can also investigate the question of whether constructs hang together as coherently when rated by generative LLMs as they do when human raters score them. To examine this, we estimated the internal consistency (Cronbach's α) of each trait scale for each LLM (see Supplementary Tables 12 and 13 for individual trait and LLM reliability results). For Sample 1, we found that LLM-generated trait scores generally had good internal consistency (average α across all traits and LLMs = 0.76; self-report average α across traits = 0.78), with some exceptions, largely in the openness domain, which had relatively lower reliability across LLMs (openness α range across LLMs = 0.26–0.79). For Sample 2 traits, we found acceptable to excellent internal consistency for all domain scales (average α across all traits and LLMs = 0.88; self-report average α across traits = 0.88), with the lowest value across all models and scales being 0.70.

Somewhat related to this, we examined how many minutes of spontaneous thoughts and how many diaries were required to achieve stability in the effect sizes. We found that at 15 min of spontaneous thoughts and 5–7 diaries, the observed effect sizes approximated the effect sizes in the full sample. Supplementary Figs. 1 and 2 detail these results.

Discussion

Across two very different samples and assessment contexts, we collected relatively brief, open-ended narrative data that we subjected to zero-shot ratings for broad personality trait domains using multiple off-the-shelf generative pre-trained transformer LLMs and associated them with participant self-reported traits. How should we understand the effect sizes of these associations? One way is to compare them with other cross-method associations in personality traits. The personality trait literature is rich with tests of agreement between self-reported and various other methods for assessing traits. For instance, Connelley and Ones[19] meta-analysed self–other agreement among personality trait ratings. They showed that trait agreement was moderated by relationship, and family and friends were generally the strongest. Our effect sizes are mostly in line with, though often exceeding, the meta-analytic family/friend ratings (see Fig. 2 for comparisons). Alternatively, we

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Table 3 | Examples of statements selected by ChatGPT 4.1 as exemplars to support its scoring

Trait	Trait level	
	High	Low
Agreeableness		
Sample 1	i want to take care of them thats all i want to do take care of people and i dont want it for people to tell me im a good person that just feels like my calling im gonna do it	ive lost my filter and i dont have any care left to like to be tactful about telling people whats wrong with something
	people smiling, happiness, happiness is the ultimate goal	i really dont care if i stay home alone for thanksgiving
	im sure once we have the pros and cons we can talk through it and come up with a good plan	i hate people from cali so fuck that especially la fuck them
Sample 2	i listened, and i asked her questions. how i felt during it. i was pretty sad for her, because it felt like it was something that she was really worried and concerned about.	i got very upset, i overreacted maybe a little bit, told him i never wanted to talk to him again, i told him to go fuck himself.
	i've just been checking in with him the whole day, most of the day. i felt mostly worried and concerned because i haven't seen him in that much pain	i behaved very impulsively and angry, and confrontational, and very accusational towards him
	i was happy to have made him happy, and to make his day better.	i definitely reacted more impulsively, i got angry and started yelling, not keeping my temper to get the things that i needed, so afterwards it kind of made me feel stupid, but i still don't like to not say what i think or feel when i feel that it needs to be said
Conscientiousness		
Sample 1	after a lot of stuff to do i have to do work im very busy today	i never leave myself enough time im always rushing its just story of my life i dont know why i do that
	im thinking about upcoming exams that i have and if im behind in classes and what im going to have to start studying	i just seem to always get myself in these cycles where im just rushing around and then im late
	usually ill make a plan in my mind	i want to be in bed
Sample 2	it was a culmination of many months of hard work, and so it felt pretty good that we were able to make some progress there	i just didn't plan accordingly, and i ended up being late to class and almost missing some participation points because of that.
	i behaved, at first i felt, or at first i was pretty calm, and i tried to collect my thoughts to get the job done	it doesn't sound like a lot, but usually i have really hard trouble doing stuff like that. chores, laundry, grocery shopping, cleaning, doing the dishes.
	s one of the things that i'm kind of a dictator about, if you want to call it that, because it drives me bonkers when stuff's a mess.	really stressed, angry at myself for not planning better, and using my time more efficiently so that i could have gotten it done earlier.
Extraversion		
Sample 1	a bunch of us in my pledge class for the business fraternity are gonna surprise him and were gonna go to karaoke tonight and that should be a lot of fun	on the bus on the way here i ran into this guy from my physics study group and he started talking to me and i was like please just go away
	joined a business frat this year and some people some of my best friends here on campus its really really changed my social life here which is fun fun	im not a very social person
	excited about my house for next year ill be living with my three best friends	i like living on my own
Sample 2	i got to hang out with some really good friends of mine. the energy was very high.	i don't like being around groups of people, so it was a bit uncomfortable for me.
	i felt good, i felt amazing.	i kind of stayed in the house all day long. i slept a lot.
	we went on a tube, like on the waves and stuff, and i've never been on a tube, so it was super fun. we were like bouncing all over the waves, and crashing back and forth	timidly, trying to just get out of the situation as quickly as possible
Neuroticism		
Sample 1	i just feel like its me against the world sometimes and it kind of sucks	i feel so privileged to even be able to afford to come here
	because sometimes i just get very sad and i dont really know why	im so relaxed maybe i should sit up
	i guess right now what im thinking about is how my life is kind of crumbling around me right now	i cant wait to cook
Sample 2	i panicked. i was really, i was crying, i was yelling, i was feeling really bad, i was anxious, i was angry, i was, oh, i was a mess.	i felt really at ease, and, like, i could just be myself
	pretty rough day, and that's about it. mood is pretty low. not feeling too good about myself right now.	i felt happy to see it, because i like it, and my boyfriend was there, and we had a good time.
	I feel useless. and helpless.	overall it was a good experience, today was good.

Table 3 (continued) | Examples of statements selected by ChatGPT 4.1 as exemplars to support its scoring

Trait	Trait level	
	High	Low
Openness		
Sample 1	everything deep and personal is basically what im thinking about all the time	just sitting here thinking of something to think to say
	we went to so many art museums	honestly the thing that bothers me the most and i know it offends people is i dont get what the hell you do with an art degree
	thinking about deepak chopra and what he has to say about reality and all that jazz	i had a bagel for breakfast this morning and a yogurt bar that was it it was pretty good
Sample 2	we stopped by an open market on our bikes, and i had this immediate connection with a woman, one of the vendors in the market	my most significant event of the day today was just getting my car inspected.
	my most significant event today was going with my boyfriend to the closing performance of the opera. that was his second opera, i've been to about ten or so, but we both agreed that it was our favorite, actually. we had really a lot of fun, and although the subject matter was not the most pleasant, we really enjoyed ourselves.	there is nothing really that eventful in my day to day. spent time with my kids, my daughter didn't have school, worked on her hair some more, watched tv, cleaned, the usual, stayed at home.
	my most significant event today was voting for hillary clinton, as i mentioned in another assessment. it felt great to be able to vote for the first major party female presidential candidate.	my day is basically simple. it's a routine. and it always remains the same.

can look at agreement between global retrospective self-report and averages of ecological momentary assessment of personality states as reported in a meta-analysis by Fleeson and Gallagher[33] (extraversion = 0.42; agreeableness = 0.55; conscientiousness = 0.48; neuroticism = 0.53; openness = 0.56). These associations are comparable in size to the convergence between self-report and LLMs that we found and are perhaps most informative as comparisons for Sample 2, which used daily video diaries. Another relevant comparison is Youyou et al.'s[9] bespoke machine learning model (MyPersonality) based on Facebook likes, which achieved similarly high agreement (extraversion = 0.45; agreeableness = 0.38; conscientiousness = 0.42; neuroticism = 0.40; openness = 0.51). In either case, our results are comparable to these benchmarks, demonstrating that zero-shot LLM scoring of very brief open-ended responses to unconstrained prompts is as good as any other cross-method approach to scoring personality.

Further, in Sample 2, we showed that the LLM-rated traits had very similar nomological networks to self-reported traits. These associations show that these LLM trait scores are not narrowly associated with their target constructs (that is, the same self-rated trait), but rather they contain much the same information contained in self-reported traits as reflected in external correlations as the self-reported traits do. The external measures used for associations here are also self-reported, so perfect profile similarity cannot be expected. By way of comparison, we have previously found that similar comparisons of the nomological nets among cross-sectional and ecological momentary assessment reported traits achieve an average profile correlation of 0.81, with a range of 0.47–0.99 (ref. 7). Thus, we see strong correlations of the patterns or profiles of associations with external measures for LLM and self-rated personality, the strength of which are in line with other cross-method comparisons of profiles. In the same sample, we also showed that LLM-scored traits can predict trait-relevant daily behaviour, psychiatric diagnoses and mental health treatment seeking. These patterns are highly consistent with meta-analytic associations between personality traits and psychiatric diagnoses (for example, [22,34]). They also show that LLM-generated trait scores are predictive of everyday, real-life behaviour as well as major psychiatric outcomes. In the aggregate, these findings show that it is possible to score complex psychological constructs from very small amounts of natural language data in a manner that is both reliable and valid.

Bridging and broadening our methods

Our results showcase a method to glean information cutting to the core of who a person is, all from brief, unconstrained, open-ended responses. Indeed, in selecting the Big-Five personality traits, we have pushed the boundaries of what generative LLMs might be able to score from a single narrative. Because these traits span affect, social functioning, cognition, and behavioural planning and organization, our findings suggest that this method can be applied to reliably and validly study many diverse, wide-ranging personality constructs and beyond. As we highlighted above, many constructs can be organized under the Big Five's domains in terms of content[21]. We think that it is reasonable to extrapolate and expect that many of these alternative constructs could similarly be scored from the same narratives. Furthermore, the fact that we can obtain ratings from LLMs that show high agreement with self-reports with (a) just a small amount of data (~15–30 min per person), (b) coming from responses to different types of prompt and (c) within two distinct study designs indicates that many different text responses might produce similarly rich scoring opportunities. Once collected, such text responses are far more valuable in many respects than traditional questionnaire data. With open-ended and unconstrained assessments, participant burden is reduced relative to lengthy surveys and there is no need to make painstaking decisions about the finite measures that you can afford to administer because the same text could be rescored again and again to measure different constructs.

A major advantage of generative LLMs is that they are inexpensive and easy to implement compared with other methods of scoring unconstrained data. To achieve these same results, using human coders would have required many months of training, coding and supervision. What would take a traditional lab months and countless person-hours of work, generative AI accomplishes in minutes. Traditional machine learning and LLMs with domain-specific additional training ('fine-tuning') address some shortcomings of human coding, but these methods require considerable AI expertise, require large training sets that limit the types of data available for training stimuli, and are otherwise resource-intensive to implement. We show that pre-trained generative LLMs produced equivalent or superior results to these state-of-the-art machine learning methods. But they did so 'off the shelf', that is, zero-shot, without the need for additional training steps, making these methods easy to implement by non-specialists. Thus, commercially available generative LLMs are not only a powerful tool to precisely annotate and score open-ended text, but they are also superior to many alternative AI options owing to their accessibility.

Most importantly, generative AI has the potential to meaningfully expand our understanding of personality by widening the aperture through which we consider information. The psychometric paradigm,

which spawned the Big Five, has been the dominant lens through which we have come to study personality in large part owing to the ease of implementation of self-report scales that offer quick and easy scoring into psychometrically robust trait values. At the same time, there is wide recognition that the self-reported Big Five are not 'all of personality'. For instance, one perspective contends that personality goes beyond traits to include the narrative self, that self-knowledge is organized as a life story[10]. A related perspective is that key parts of personality are idiosyncratic, and a full understanding of an individual requires idiographic studies that captures their uniqueness[3,35,36]. While there is growing interest in these approaches, they have met with very real challenges associated with difficulty of data collection and costs of scoring. Our results illuminate a path forward by showing that nomothetic constructs that are presumed to apply to all to some degree, such as the Big-Five traits, can be reliably and validly scored from idiographic narratives generated in someone's own words. Moreover, these are quotidian reports—the same everyday experiences that define much of who we are. Bridging this traditional divide facilitates a more thorough study of personality that is ecologically valid and highly relevant to individuals. Yet, this is just the necessary first step, opening a window into additional and novel approaches that might come from the ability to rapidly and accurately score unconstrained narratives.

Opportunities and future directions

Our study points to opportunities to incorporate generative LLMs widely in personality research and psychological sciences more broadly. We have taken a critical first step, showing that impressively reliable and valid scoring is achievable for broad psychological content using cheap and ubiquitously available generative AI. Measurement matters; it is the bedrock of all scientific enterprises. We can now more confidently proceed in using generative LLMs to interrogate these types of data in other ways. The most obvious are to include open-ended assessments in studies alongside traditional scales, or in repeated measures over time. This could dramatically increase the breadth of assessments, easily allow for multimethod data or facilitate the assessment of concepts for which there are no extant self-reported scales or that might elude self-report (for example, aspects of self and identity functioning[37]). Also, generative LLMs could be enlisted to investigate the very nature of psychological concepts, such as which aspects of constructs are most definitional, or what the causal logic is among components of constructs. Owing to the method's ability to accommodate idiosyncrasies of people's experience, it might also be enlisted to refine concepts and their measurement or recognize similar constructs that might be reflected in very different behaviour or narrative reports across people or across context within persons. It might also be useful in developmental research to study continuity and change over time, even as individuals change in the style and features of their reports, which has traditionally challenged research (that is, the measurement invariance problem).

Leveraging generative LLMs will require tailoring the method for different applications. One consideration is that the bandwidth-fidelity trade-off operates here in the solicitation of open-ended data, just as it does in traditional psychological scale development. Some research may benefit from a widely open aperture, using broad prompts, little direction and minimizing constraints on the resulting narratives. This will be ideal for exploratory work, or work that seeks to understand the individual holistically, as is the case in personality science. Yet, other scenarios might call for a narrowing of the aperture, filtering the resulting narrative into a more focal stream. For instance, research that wants to ensure very detailed coverage of specific phenomena or where determining the presence or absence of a construct is required to make an inference (see depression studies reviewed above). Less constrained prompts will allow for scoring more constructs, but more constrained prompts might allow for more detailed scoring and greater fidelity and confidence in the scoring. Investigators will need to consider their goals when designing assessments of qualitative data.

Indeed, participant prompt engineering is a crucial design choice, and the implications and issues related to it are a key future direction and opportunity for methodological research. For instance, how well do broad, unconstrained prompts do at eliciting the breadth of relevant information (for example, in a baseline assessment prompt: 'Please describe your personality fully'; a daily diary prompt: 'Please summarize your day') or how much additional information beyond what is queried is contained within more specific prompts (for example, how much affective, cognitive and motivational information is contained in a specific social prompt such as 'Describe how you typically behave with your spouse')? Future work may even benefit from active follow-up or adaptive prompting that starts broad (for example, describe your personality) and then follows up with specific prompts if specific information is lacking (for example, how do you typically behave at work). Or they could be compared directly. The space of potentially interesting directions is vast. But the takeaway is that there should be methodological research devoted to this form of assessment in the same way that it happened with previous paradigms, such as the psychometric approach.

Where and how the data are collected may ultimately matter a great deal. There may be more nuanced contextual differences that contribute to results that one might expect to achieve. For example, we found markedly better performance of zero-shot LLM ratings using brief prompts than did Peters and Matz[15] when applying a similar method to Facebook posts (~33% better performance). In addition, in our own two studies, we found differences in performance as it related to predicting some traits. Take how conscientiousness and agreeableness associations were much stronger in Study 1 (spontaneous stream of thoughts) compared with Study 2 (most significant event), whereas the extraversion association was stronger in Study 2; in fact, it was the strongest. It may be that sitting quietly in a room pulls for more reflection on upcoming tasks, commitments and one's social duties than do daily significant events, which may elicit reports of more outgoing and exciting social experiences and behaviour. This interpretation is speculative at this juncture, but we raise it to motivate deeper thinking about the next generation of study designs and analyses.

We also did not engage in LLM-prompt engineering. The exact prompts that we used are presented in Methods. Our choice to have LLM's rate constructs at the item level, as opposed to giving the LLM the construct name or a detailed definition as was done in Peters and Matz[15], ultimately performed well. Indeed, this is a major distinction between our and their work. Furthermore, item-level assessment allowed us to assess the internal consistency of the LLM ratings and could be extended to additional psychometric inquiries. Item-level assessment also allows maximum extensibility to new psychological constructs because it shows that LLMs can assess any construct with an associated scale from open-ended subject data. Nevertheless, like with participant prompts, systematically investigating the impact of alternative LLM prompts on the elicited information is an important area for future inquiry. For example, what, if any, difference emerges when comparing item-level as opposed to construct-level prompts? There is a large practical difference, in that a prompt that rates each item independently requires running many more LLM ratings than a prompt that asks for one rating of a whole construct and therefore is much more costly monetarily. If the latter were shown to be non-inferior, it would provide marked efficiencies in cost. As another example, here we ran a separate LLM conversation per item, but one could request that the model rate items in sequence within the same conversation. In some respects, this might be a close match to how human participants complete questionnaires, as a continuous stream where the memories of responses to previous questions might influence responses for subsequent questions. This is another exciting and important area for future inquiry.

Where we go next with this method depends on how we think about generative LLM ratings of personality. The personality literature stands

out for its in-depth consideration of the validity and value of different reporters' perspectives and their agreement, as well as meaningful differences across assessment modalities. Our achieved convergence correlations between LLMs and self-report were compared with self–other agreement benchmarks (family and friends), ecological momentary assessment rated behaviour, clinical interviews and existing machine learning algorithms. Any time one compares scores across methods or raters, convergence correlations are attenuated, sometimes drastically so. In addition, self and other ratings of traits are understood to be blends of multiple influences, being composed of 'true' variance as well as self-concept, impression management and observability (that is, internal versus external expression of traits) related variance. Therefore, perfect agreement between self-ratings of traits and another rater or method is neither expected nor desirable. Therefore, in the context of multimethod associations, our achieved effect sizes are strong.

In this respect, generative LLM ratings could be considered as another perspective, another reporter, or they could be refined to maximize agreement with self-reports. The last option may be the one that first comes to mind to many readers. The associations that we find are very good, but there is room for improvement. Understandably, thinking of someone's own rating as the ground truth makes a lot of sense and would be the most direct bridge between narratives and self-report scales. It would have high practical import, as we have described above. Although we are impressed by generative LLMs' ability to score complex traits in high agreement with self-report, we stress that these were not assessments designed to maximize the associations with self-report ratings of personality. Much higher agreement with self-reported traits is likely achievable with prompts engineered specifically to elicit one's own self-concept and general patterns of behaviour of the sort captured by trait measures. That those used here are based on largely unconstrained data makes the observed associations even more remarkable and may reflect lower bound effect sizes that could be increased with more focused prompts. Thus, the effect sizes should be interpreted with this in mind. These were brief (quasi) naturalistic verbal diaries of meaningful events over a half hours' worth of mind wandering (Sample 1) or one event per day over about 2 weeks (Sample 2). Even with the current level of agreement, we see broad applicability to existing narrative data of various sorts, including those that may have been collected for research studies as well as any large public repository of an individual's narrative or diary data (for example, public figures).

However, it may be that generative LLMs may serve as a vehicle to expand the assessment of personality, offering a different perspective, and therefore should not be expected to have perfect associations with self-report. The contemporary view is that there is no ground truth, rather that the disagreement between raters of personality (for example, self- versus informant report) is based on an asymmetry of knowledge[38,39], and that there are validity and interesting variance in the discrepancies, not just error. For instance, self- and other-rated personality often increment each other in the prediction of important life outcomes[20,38]. Generative LLM scoring of personality may offer a similarly distinct and valuable approach to understanding personality. The current datasets did not offer sufficient non-self-reported data to adequately test this possibility because self-reported psychological variables were largely what we had access to and these do not offer a fair comparison owing to overlapping method variance, but we see it as a compelling new direction.

An exciting question is how we should conceptualize what generative LLMs 'know' about personality. It may be a distilled measure of social consensus, folk concepts, or societal/cultural expectations or constructs. Therefore, by comparing generative LLM scores with individuals' own ratings, we could identify ways in which an individual's experience differs from how others might see it. This has clinical and applied implications, where discrepancies in perception and construal are often presumed to be the source of interpersonal difficulties[40].

Limitations and other considerations

All studies have limitations. Among the most salient here is that our samples are not representative of all samples that we might want to study using generative LLM ratings of text. Although our samples had a range of ages, both had mean ages relatively early in adulthood (24.5 and 28.0 years, respectively) without strong representation from middle- and older-aged adults. Sample 2 was roughly balanced on sex, although Sample 1 had a larger portion of women than men. As a proportion of the samples, some races often under-represented in research were, in fact, well represented. But representation of participants of races other than White were small in terms of absolute numbers. Ultimately, we did not have large enough samples to compare performance across groups for which performance might differ. Therefore, we did not perform any exploratory analyses on age, sex, race or ethnicity. A wealth of research has shown that trained AI models are prone to biases in their training sets that may under-represent some demographics, including age, race, ethnicity or sex. To make this abstract point concrete, to the extent that different types of individual systematically express language differently, it would potentially undercut the generalizability of the methods that we used here. As it relates to scoring personality from open-ended text using commercial generative LLMs, this remains an unstudied issue that future research should address. The issue can be conceptualized in terms of measurement invariance: across key social and demographic groups, are individuals with similar levels of a trait ascribed the same (or similar) values by the LLM based on the language they use?

An additional limitation concerns interpretability. We highlighted how the models can be used to generate 'explanations' either in the form of rationales or quotes (Table 3) for why they scored text the way they did. This is a useful technique that goes some way towards showing that the coding is sensical and that the model 'understands' the relevant construct. Nonetheless, model-generated explanations and quotes are post hoc and do not provide a direct summary of the model internal reasoning. In this regard, they are conceptually similar to post hoc explainable AI techniques such as Shapley Additive or Local Interpretable Model-agnostic Explanations. Therefore, they should not be interpreted as necessarily reflecting the actual processes used by the LLMs in producing their results. Such process-level understanding can only be gained through mechanistic interpretability methods[41,42], and application of these increasingly powerful methods to ascertain how LLMs decode psychological constructs from unconstrained text should be taken up in future research. In this vein, it would also be important to further compare LLM ratings of text with human ratings of the same text (for example, ref. 19) to evaluate points of convergence and divergence there.

We observed that different LLMs arrive at slightly different scorings of open text. In some cases, the differences are modest (for example, neuroticism scores in Sample 2), but in other's they are quite pronounced (for example, openness in Sample 1). Interestingly, though, there was no single model that consistently performed the best or worst. We also demonstrated that using the average score across LLMs results in higher convergence with self-reports than many individual LLMs. It is notable that each LLM is trained on somewhat different training data and they are given different 'tuning' during post-training to follow instructions and produce human-preferred outputs. Thus, it is reasonable to think of each LLM as a different rater, each with a slightly different 'perspective', and, for standard psychometric reasons, the shared perspective is more accurate than any single perspective. Thus, it may be best to use multiple LLM 'raters' when scoring open-ended text. Indeed, this may be one reason we achieve better overall agreement with self-report than did Peters and Matz[15]. Alternatively, with more systematic research accumulating, one model may prove to be generally better or, more likely, some models may have advantages for some coding tasks. At this juncture, we believe that it would be premature to conclude that particular models have areas of relative

strength or weakness in scoring specific types of construct, but this may be something to attend to as this method is applied more broadly. Accordingly, for now, our recommendation is to use multiple LLMs for similar rating tasks until there is sufficient accumulated data to suggest reliable differences among models or that a single model consistently does substantially better. There is not yet enough information to justify another approach.

Finally, the present study used participants' transcribed thoughts and reflections that were non-clinical and non-sensitive in nature. Given the abilities showcased by LLMs in quantifying important psychological constructs from this type of data, a natural next step is to extend the approach to the clinical setting or other settings involving sensitive data—for instance, having participants explicitly describe daily symptoms, substance use, illegal behaviour or even suicidal ideation. The use of commercial LLMs with protected health information or other sensitive information raises several privacy and data protection considerations. These include ensuring compliance with relevant regulations such as the Health Insurance Portability and Accountability Act and the General Data Protection Regulation as well as institution-specific restrictions; implementing strong safeguards around data storage and transmission; and maintaining transparency with participants about how their data will be used. In addition, researchers should minimize risk of re-identification of subjects, even in datasets ostensibly with minimal personal identifying information. For very large datasets, recent advancements in automated de-identification (with semantic preservation) with locally implemented, open-source LLMs are an attractive option (for example, ref. 43), as these methods perform de-identification entirely locally before any third-party sharing.

Summary

In sum, these results show that inexpensive and widely available pre-trained transformers provide an unprecedented tool for personality science to quantitatively assess psychological constructs in open-ended qualitative data sources. We demonstrated this in two very different types of open-ended narrative with low constraints on responses. Moreover, our results also serve as yet another reminder of personality's ubiquity, in that it is carried in the stream of our resting thoughts and is woven into the fabric of our daily lives. Perhaps most importantly, the approach that we describe in this study opens a pathway for widespread adoption for researchers across the psychological and social sciences. Open-ended voice diaries are inexpensive to collect, impose minimal subject burdens and—as we demonstrate in this study—contain a wealth of information decodable by LLMs about important psychological constructs, without any of these constructs being specifically prompted from subjects. Thus, the present study provides a clear and concrete model of a research approach that can be readily adopted widely by researchers in their own personal studies and projects, large and small. The next steps will be to enlist this tool for mechanistic and predictive efforts. These results should encourage behavioural scientists to collect more unconstrained, open-ended data or mine existing datasets—be they narratives, diaries, transcripts of social exchanges in the lab and so forth—all of which are amenable to rescoring with LLMs. We encourage others to continue the methodological testing of LLMs as well as their use to study constructs of interest.

Methods

This research complies with all relevant ethical regulations and was approved by the University of Michigan Institutional Review Board (Sample 1, protocol number HUM00134688) and the University of Pittsburgh Institutional Review Board (Sample 2, protocol number PRO15120131).

Sample 1: spontaneous stream of thoughts sample

Subjects consisted of two samples that underwent identical lab procedures. Sample 1 was recruited in 2018 through flyers placed on campus at a large public university and a university-sponsored participant recruitment website, as well as recruited from an undergraduate subject pool at the same university in fall term 2018. The overall included sample (N = 60) ranged in age from 18 to 67 (M = 24.5, SD = 12.2) and was 63.3% female. Most participants identified as White (60.0%, n = 36), 11.67% identified as Black or African American (n = 7), 20.0% as Asian (n = 12), and 8.3% (n = 5) as multiracial or 'other'.

All participants provided informed content. Participants were given a brief explanation of the stream of spontaneous thought using examples of when this type of thought arises. For example, they were told that these are the kinds of thought that would arise 'during a long bus ride when it is dark, you are wide awake, and no one else is around'. Subjects were allowed to ask questions and receive clarifications until they assented that they understood what they would be doing.

Subjects then sat alone in a small fully enclosed room with minimal external stimulation (quiet, dimmed lights, no books on shelves, no items on the wall and limited foot traffic near the room). Their voices could not be heard by anyone outside the room. They verbalized their spontaneous thoughts for 30 min. A digital voice recorder was placed on a nearby table and recorded their speech.

Sample 1 measure

The NEO Five-Factor Inventory (NEO-FFI[27]) is a 60-item self-report measure designed to assess the Big-Five personality factors: neuroticism, extraversion, openness to experience, agreeableness and conscientiousness. Derived from the longer 240-item NEO Personality Inventory-Revised (NEO PI-R), the NEO-FFI provides a brief yet reliable assessment of these domains. Each domain is measured by 12 items selected from the original NEO PI-R based on their item-total correlations and content validity. Respondents rate items on a five-point Likert scale ranging from 'strongly disagree' to 'strongly agree'.

Sample 2: daily diary sample

In Sample 2, community members (N = 311) were recruited between 2016 and 2018, both online and through posted flyers for the ambulatory assessment, personality, ecological context and stress (AAPECS) study. Only the first 108 participants completed nightly video diaries and are included in this report. The diaries were subsequently discontinued from the study to make room for other study procedures among subsequent participants. In the larger sample, screening was used to recruit a sex-balanced sample that was also selected to balance individuals who had received recent mental health treatment (within the past year) with those who had not. Participants were also screened to have a range of psychopathology.

The included sample (N = 108) ranged in age from 18 to 40 (M = 28.0, SD = 6.1) and was 53% female. Most participants identified as White (78%, n = 84), 17% identified as Black or African American (n = 18), and the remaining 5% (n = 6) identified as another race. Of the sample, 69% had a lifetime history of mental health treatment, of which 66% were currently receiving treatment at baseline.

All participants provided informed content. Participants completed a lab session and an ambulatory assessment protocol. The length of the ambulatory assessment protocol was 21 days for the initial 37 participants but was then reduced to 14 days for the remainder owing to concerns about burden. We use the baseline self-report inventories and the daily video diary portions of the ambulatory assessment.

For the ambulatory assessments, participants typically used their own Android or iOS smartphone. Laboratory-owned Android smartphones were available, allowing individuals without compatible devices to participate.

All participants received US$50 for the baseline session. Those who answered 90% or greater of the surveys during the ambulatory assessment period earned an additional US$110 (14-day ambulatory assessment) or US$160 (21-day ambulatory assessment). This amount was prorated by week for those who completed less than 90% of the surveys overall, with surveys completed in later weeks of the study valued

higher in the prorated compensation model. In addition, for each 100 participants, a drawing was conducted for an iPad Mini. Chances of winning increased with the number of surveys answered.

Measures

Participants completed a demographic questionnaire including items addressing sex, age, ethnicity, race, substance use, and previous or present mental health treatment.

IPIP-NEO. The IPIP-NEO[29] is a short-form, open-source alternative to the flagship assessment of the Five-Factor model of personality—the Revised NEO Personality Inventory (NEO PI-R[27]). The IPIP-NEO-120 uses 120 items to provide estimates of 5 NEO domains (openness, conscientiousness, extraversion, agreeableness and neuroticism) and 30 subdomains (6 per domain). Items (for example, 'I get angry easily') are rated on a 5-point Likert-style scale from 0 (very inaccurate) to 4 (very accurate). Domain-level omegas in this study were 0.83 for agreeableness, 0.84 for openness, 0.88 for extraversion, 0.91 for neuroticism and 0.92 for conscientiousness.

Comprehensive Assessment of Traits Relevant to Personality Disorder (CAT-PD). The CAT-PD[44] includes 216 items and assesses 33 maladaptive personality traits with self-report items rated on a 5-point Likert scale (0, very untrue of me; 1, moderately untrue of me; 2, neither true nor untrue of me; 3, moderately true of me; 4, very true of me). Reliability for all scales was McDonald's $\omega > 0.79$.

Inventory of Interpersonal Problems-Short Circumplex. The Inventory of Interpersonal Problems-Short Circumplex[45] is a 32-item measure that measures difficulties in interpersonal functioning with items (for example, 'I am too easily persuaded by other people') rated from 1 (not at all) to 5 (extremely). Scores are provided for each octant as well as for dimensions of dominance and affiliation.

Structured Clinical Interview for DSM-5 Disorders–Clinician Version (SCID-5-CV). Portions of the *Structured Clinical Interview for DSM-5 Disorders: Clinician Version* (SCID-5-CV[46]) were administered in this study to assess internalizing disorders (current and past mood and anxiety disorders). Disorder criteria were rated as either absent or present. Based on SCID-5 scores, 26% of the sample met criteria for any internalizing disorder, including generalized anxiety disorder, current major depressive episode, obsessive compulsive disorder, social anxiety disorder, panic disorder, agoraphobia or post-traumatic stress disorder. Internalizing diagnoses were scored as either present or absent for this study.

Structured Interview for Diagnostic and Statistical Manual of Mental Disorders, Personality Disorders (SIDP-IV). The *Structured Interview for Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition, Personality Disorders* (SIDP-IV[47]) assesses criteria for the ten DSM-IV personality disorders. The format is semi-structured and arranges questions regarding specific diagnostic elements in a thematic manner (for example, 'close relationships' and 'perception of others') rather than by discrete disorder. Interviewers rated participants as to the degree of presence of all criteria, following specific scoring notes. For one section, 'observational criteria', interviewers made ratings regarding participant behaviour, demeanour and dress. For all criteria, a participant scored 0 for 'not present', 1 if 'sub-threshold' manifestations were apparent, 2 for 'present' and 3 for 'strongly present'. Timeframe, unless otherwise specified, was limited to the past 5 years (or since age 18 for younger participants). All personality scores were averaged for this study to create a general measure of personality disorder.

Daily video diaries. Daily diary assessments typically began the day after baseline assessment. Surveys were administered through the MetricWire smartphone application, and participants were prompted to answer surveys through push notifications. Participants were prompted to complete a daily survey and the video diary, which was recorded on their phone, around their typical bedtime and were given 3 h to respond. Brief (1 min) video diaries were collected, and participants were instructed to describe the most significant event that occurred during the day, including information about what happened, how they felt and behaved during the event, and who else (if anyone) was present. Responses were recorded with the smartphone's self-facing camera and submitted using the MetricWire application.

Ecological momentary assessments also typically began the day after baseline assessment. The length of the assessment period was initially set at 21 days but was later shortened to 14 days. Surveys were administered through the MetricWire smartphone application, and participants were prompted to answer surveys through push notifications. Surveys were delivered on a randomly initiated schedule. Participants were delivered 6 surveys per day during an approximately 12 h time window that corresponded to the participants' typical waking hours. Blocked random intervals were set so that a minimum of 90 min passed between surveys and participants were given 20 min to initiate a response to each one. A mean number of 68.62 (SD = 20.94) surveys were completed per participant.

Participants received ambulatory assessments focusing on momentary measures of affect and stress with a focus on interpersonal interactions since the last assessment. Participants first answered items assessing current positive and negative affect. Next, participants were asked if anything stressful had occurred since the last assessment. Stress was defined broadly, and participants were instructed to consider whether they themselves felt stressed when the event occurred, not whether they or another person would typically feel stressed in the same situation. If participants reported that the stress involved an interpersonal interaction, they reported on their interpersonal behaviour during the situation. Ratings of stress severity were collected on a scale from not at all (0) to extremely (100) stressful.

Positive and negative affect were measured using ten adjectives in the randomly prompted surveys. Positive affect was represented by happy, proud, content, excited and relaxed. Negative affect was represented by ashamed, nervous, hostile, sad and angry. Participants were asked questions in the form 'How ADJECTIVE do you feel right now?'. Ratings were made on a scale from 0 (not at all) to 100 (extremely).

The Visual Interpersonal Analogue Scale (VIAS[48]) was developed to efficiently assess dominant and affiliative behaviour during social interactions. Dominant behaviour was assessed using a visual analogue slider bar ranging from −50 (accommodating/submissive/timid) to 50 (assertive/dominant/controlling). Affiliative behaviour was rated on a similar visual slider bar ranging from −50 (cold/distant/hostile) to 50 (warm/friendly/caring).

Momentary ratings were averaged across all observations to derive individual scores for use in this study.

Trait scoring procedure and LLM prompts

Both samples followed a nearly identical procedure for scoring personality traits from open-ended text using generative LLMs. The first step involved transcribing the recordings into text. Audio recordings from Sample 1 were transcribed directly, while video recordings from Sample 2 were first converted to audio-only .mp3 files using ffmpeg, an open-source multimedia processing tool. All audio files were then transcribed using the pre-trained Whisper Large model, an automatic speech recognition (ASR) model developed by OpenAI. The raw transcription was used in all scoring without pre-processing or cleaning steps (for example, adding punctuation), because modern LLMs do not need this step to process text.

To score personality traits from text, we used seven state-of-the-art, commercially available LLMs. These were OpenAI's GPT-4.1 and GPT-4.1-Mini, Anthropic's Claude-3.7-Sonnet, Google's Gemini-2.5-Flash-Thinking, xAI's Grok-3-Beta, Meta's Llama-4-Maverick and Alibaba's Qwen3-235B. Among these, Llama-4-Maverick and Qwen3-235B

are open-source models, while the remaining five are proprietary. Each model was accessed through a unified Python API, leveraging the openai package and OpenRouter API for dynamic model routing. All calls were made with temperature = 0 and seed = 0. The only exception was Grok-3-Beta, which does not accept seed = 0; in this case, the seed was set to 1. For models supporting adjustable reasoning parameters, the reasoning effort was set to 'medium'. Note that, even with zero temperature and a seed set, LLMs can still exhibit small variability across runs owing to floating point rounding, routing requests to different servers and other subtle issues. To assess this variability, we repeated our analysis with GPT 4.0 across 5 runs with identical prompts, zero temperature, and with a seed set as well as across 5 runs with identical prompt, zero temperature and seed random. In each case, we found minimal variability across these runs and report these results in Supplementary Tables 14 and 15.

Our approach was to frame each questionnaire item from the same personality inventory that the participant completed at baseline (NEO-FFI or IPIP-NEO-120) as a structured inference task for the LLMs. Each prompt instructed the LLM to respond to an individual item as if it were the participant, relying on the full transcript of either spontaneous thought (Sample 1) or daily diary (Sample 2) as context. The models were explicitly instructed to select from the same fixed set of Likert-style response options used in the self-report versions of the inventories. Here are some example items from the IPIP-NEO-120: I worry about things (neuroticism), I feel comfortable around people (extraversion), I have a vivid imagination (openness), I love to help others (agreeableness), and I like to tidy up (conscientiousness). The full set is available at https://ipip.ori.org/30FacetNEO-PI-RItems.htm. The NEO-FFI's copyright does not allow printing the items in scholarly publications, but they are of a very similar nature to these. If a model returned a response that did not exactly match one of the allowed options, the response was discarded, and the prompt was re-issued until a valid response was returned. A separate prompt was generated in a new conversation for each item for each participant until all of the items were scored.

Here is the exact prompt presented to LLMs for Sample 1:

'Your task is to respond to the following NEO-FFI question based on the participant's spontaneous stream of thoughts, provided below. Respond as though you are the individual who generated these thoughts, reflecting their personality traits.

Base your answer on inferred personality traits. Think carefully about what the thoughts imply about tendencies and behaviors.

For each question, select the most appropriate option:

- Strongly Disagree: The statement is definitely false or the participant would strongly disagree with it.
- Disagree: The statement is mostly false or the participant would generally disagree with it.
- Neutral: The participant would be neutral on the statement, cannot decide, or find the statement equally true and false.
- Agree: The statement is mostly true or the participant would generally agree with it.
- Strongly Agree: The statement is definitely true or the participant would strongly agree with it'.

NEO-FFI question to answer:
[[NEO-FFI item]]
Participant's spontaneous stream of thoughts:
[[Full Transcript]]
Your response must be exactly one of:
Strongly Disagree
Disagree
Neutral
Agree
Strongly Agree

Do not include any explanation, punctuation, or additional text. Return only the exact phrase from the list above.

Here is the exact prompt presented to LLMs for Sample 2:

'Your task is to respond to the following IPIP-NEO-120 question based on the participant's daily diaries of the most significant event that occurred during the day, provided below. Respond as though you are the individual who generated these thoughts, reflecting their personality traits.

Base your answer on inferred personality traits. Think carefully about what the thoughts imply about tendencies and behaviors.

For each question, select the most appropriate option:

- Very Inaccurate: The statement is definitely false or the participant would strongly disagree with it.
- Moderately Inaccurate: The statement is mostly false or the participant would generally disagree with it.
- Neither Accurate nor Inaccurate: The participant would be neutral on the statement, cannot decide, or find the statement equally true and false.
- Moderately Accurate: The statement is mostly true or the participant would generally agree with it.
- Very Accurate: The statement is definitely true or the participant would strongly agree with it'.

IPIP-NEO-120 question to answer:
[[IPIP-NEO-120 item]]
Participant's daily diaries:
[[Full Transcript]]
Your response must be exactly one of:
Very Inaccurate
Moderately Inaccurate
Neither Accurate nor Inaccurate
Moderately Accurate
Very Accurate

Do not include any explanation, punctuation, or additional text. Return only the exact phrase from the list above.

The full transcript and questionnaire item were included with these prompts. A separate prompt was generated in a new conversation for each item for each participant until all of the items were scored. Each model's selected answer was recorded and converted to the corresponding numerical score using the scoring key of the relevant inventory. This process was repeated for each item, generating a complete set of item-level LLM-inferred scores for each participant.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The raw self-report measures data and data generated by LLMs used in the data analysis have been compiled and deposited via GitHub at https://github.com/SripadaLab/personality_llm_zero_shot/. Recordings and full text transcripts (beyond what is contained in the tables) are not shared openly, but can be accessed by contacting the authors.

Code availability

The code for analyses is available via GitHub at https://github.com/SripadaLab/personality_llm_zero_shot/.

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Author contributions

All authors contributed conceptually to the design of this study, A.G.C.W. and C.S. designed and funded the data collection, statistical analyses were conducted by A.T., M.A. and A.G.C.W., A.G.C.W. drafted the paper, and all authors edited it.

Competing interests

The authors declare no competing interests.

Additional information

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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	To score personality traits from text, we employed seven state-of-the-art, commercially available LLMs. These were: OpenAI's GPT-4.1 and GPT-4.1-Mini, Anthropic's Claude-3.7-Sonnet, Google's Gemini-2.5-Flash-Thinking, xAI's Grok-3-Beta, Meta's Llama-4-Maverick and Alibaba's Qwen3-235B. Among these, Llama-4-Maverick and Qwen3-235B are open-source models, while the remaining five are proprietary. Each model was accessed through a unified Python API, leveraging the openai package and OpenRouter API for dynamic model routing. All calls were made with temperature=0 and seed=0. The only exception was Grok-3-Beta, which does not accept seed=0; in this case, the seed was set to 1. For models supporting adjustable reasoning parameters, the reasoning effort was set to "medium".
Data analysis	All code used in the data analysis has been compiled and deposited on GitHub: https://github.com/SripadaLab/personality_llm_zero_shot/

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We make openly available the raw self-reported data and data generated by large language models in an on-line repository (https://osf.io/546ke/). Raw recordings and text transcripts are not shared.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	We report the sex of each sample's participants. Sample 1 is 63.3% female, Sample 2 is 53% female.
Reporting on race, ethnicity, or other socially relevant groupings	For Sample 1 we report, "Most participants identified as White (60.0%, n = 36). 11.67% identified as Black or African American (n = 7), 20.0% as Asian (n = 12), and 8.3% (n = 5) as multiracial or "Other"." And in Sample 2 we report, "Most participants identified as White (78%, n = 84). 17% identified as Black or African American (n = 18), and the remaining 5% (n=6) identified as another race. Of the sample 69% had a lifetime history of mental health treatment, of which 66%% were currently receiving treatment at baseline. " We also report on the age of each sample.
Population characteristics	We completed the behavioral and social sciences study design section below.
Recruitment	Sample 1 was recruited in 2018 through flyers placed on campus at a large public university and a university-sponsored participant recruitment website, as well as recruited from an undergraduate subject pool at the same university in Fall term 2018. In Sample 2, community members (N=311) were recruited between 2016 and 2018, both online and through posted flyers for the Ambulatory Assessment, Personality, Ecological Context, and Stress (AAPECS) Study..
Ethics oversight	Sample 1 was approved by the University of Michigan Institutional Review Board (HUM00134688). Sample 2 was approved by the University of Pittsburgh Institutional Review Board (

Note that full information on the approval of the study protocol must also be provided in the manuscript.

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Behavioural & social sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	This is a quantitative study. We use large language models to take open-ended qualitative data and score it quantitatively. This is the focus of the research report and we describe this extensive.
Research sample	Both study 1 and study 2 use previously existing data sets. Sample makeups are described in the report as follows: "Subjects consisted of two samples that underwent identical lab procedures. Sample 1 was recruited in 2018 through flyers placed on campus at a large public university and a university-sponsored participant recruitment website, as well as recruited from an undergraduate subject pool at the same university in Fall term 2018.", "Community members (N=311) were recruited between 2016 and 2018, both online and through posted flyers for the Ambulatory Assessment, Personality, Ecological Context, and Stress (AAPECS) Study. Only the first 108 participants completed nightly video diaries and are included in this report. The diaries were subsequently discontinued from the study to make room for other study procedures among subsequent participants. In the larger sample, screening was used to recruit a sex-balanced sample that was also selected to balance individuals who had received recent mental health treatment (within the past year) with those who had not. Participants were also screened to have a range of psychopathology. "
Sampling strategy	Please see the response to the last question.
Data collection	For Sample 1 we report, "Subjects then sat alone in a small fully enclosed room with minimal external stimulation (quiet, dimmed lights, no books on shelves, no items on the wall, limited foot traffic near the room). Their voices could not be heard by anyone outside the room. They verbalized their spontaneous thoughts for 30 minutes. A digital voice recorder was placed on a nearby table and recorded their speech." And, "Participants completed a lab session and an ambulatory assessment protocol. The length of the

	ambulatory assessment protocol was 21 days for the initial 37 participants but was then reduced to 14 days for the remainder due to concerns about burden. We use the baseline self-report inventories and the daily video diary portions of the ambulatory assessment. For the ambulatory assessments, participants typically used their own Android or iOS smartphone. Laboratory-owned Android smartphones were available, allowing individuals without compatible devices to participate. "
Timing	Sample 1 was recruited in 2018. Sample 2 was recruited from 2016-2018.
Data exclusions	No data were excluded, we used all available data.
Non-participation	All participants are included. In study, which includes an ecological momentary assessment protocol, we report on the number of surveys completed by the participants. However, the reasons for missing individual surveys in an ecological momentary assessment protocol are typically many and varied and not easily ascertained. Therefore we report ratesparticipation but not reasons for non-completion of individual surveys/diaries. We do report the relationship between number of survey's completed and effects in the supplementary materials online on OSF.
Randomization	Randomization is not applicable to this reserach.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	n/a
Novel plant genotypes	n/a
Authentication	n/a